

$$(E): y'(x) + y(x) = x$$

$$\begin{array}{l} 1. \quad \left. \begin{array}{l} f(x) = mx + p \\ f'(x) = m \end{array} \right\} \begin{array}{l} f'(x) + f(x) = m + mx + p \\ \quad \quad \quad = mx + (m+p) \end{array} \end{array}$$

donc $m = 1$ et $p = -1$ donc $\boxed{f_p(x) = x - 1}$

2. $(E_0): y' + y = 0$ donc $\boxed{f_0(x) = k e^{-x}}$ $k \in \mathbb{R}$

3. Solutions de (E) : $f_k(x) = f_p(x) + f_0(x) = k e^{-x} + x - 1$

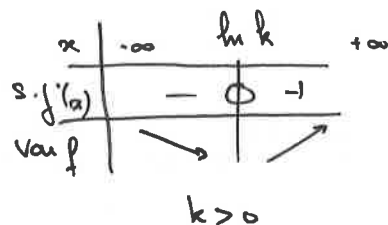
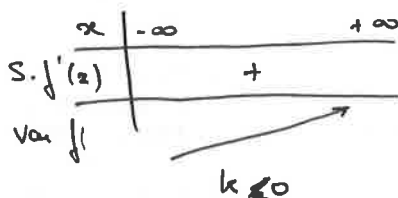
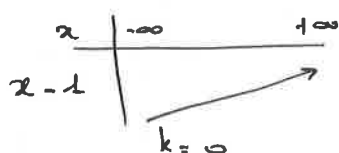
4. $f_{k_0}(0) = k - 1$ donc $k = 2$ $\boxed{f_{k_0}(x) = 2e^{-x} + x - 1}$

8. $f'_k(x) = -k e^{-x} + 1 \geq 0$ si $-k e^{-x} + 1 \geq 0 \Leftrightarrow e^{-x} \leq \frac{1}{k}$

$\Leftrightarrow -x \leq -\ln k$ si $k > 0$

$\Leftrightarrow x \geq \ln k$

Si $k \leq 0$ alors $f'_k(x) \geq 0$.



10. $f_k(x) - (x-1) = k e^{-x} > 0$ si $k > 0$ $\Leftrightarrow$ au dessus de Δ
 < 0 si $k < 0$ $\Leftrightarrow$ au dessous de Δ .

12. $f_k(0) = k - 1$ 1 seule courbe $k = 2$

13. Si $k > 0$ alors minimum pour $x = \ln k$

15. Minimum atteint: $x = \ln k$ et $y = k e^{-\ln k} + \ln k - 1$
 $= -k^2 + \ln k - 1$
 $= x - k^2 - 1 \Rightarrow$ Dérivée.